



Assignment 1: Project Scope and Plan

Team 9 · CAM_DS_401_P126 Employer Project

Employer Partner: Bank of England · Firms in scope: HSBC and Barclays

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Contents

1. Background and problem statement.....	2
2. Team roles and ways of working.....	2
3. Project scope and plan.....	3
4. Approach and work plan.....	5
Appendix A: Project roadmap.....	7
Appendix B: Work plan and measurable milestones.....	8
Appendix C: Risk register.....	10
Appendix D: Roles and responsibilities breakdown.....	11
Appendix E: Team's PREACH collaboration rules.....	12

1. Background and problem statement

The Prudential Regulation Authority (PRA) supervises ~1,500 financial firms, drawing primarily on structured regulatory returns submitted through the Bank of England's (BoE) reporting pipeline and validated against thousands of defined data points. Quarterly results announcements sit outside that pipeline. They are unstructured, they combine a curated management presentation with an unscripted analyst question-and-answer (Q&A) session, and interpreting them demands financial and technical expertise. As they are expensive to read at scale, whatever supervisory signal they contain isn't systematically used. Our problem is therefore *not* whether earnings calls can be summarised, but *whether they carry information that the reported figures don't*.

Problem Statement:

“Do quarterly earnings announcements and analyst Q&A carry a leading signal about a firm's prudential condition that reported financial metrics alone don't capture?”

2. Team roles and ways of working

Our team of seven is organised around the stages of the analytical pipeline, with one named owner for every stage and named contributors where a stage spans two roles; refer to *Appendix D* for the roles & responsibilities breakdown and *Appendix E* for the team's PREACH collaboration rules.

Communication Mechanism	Purpose & Approach
Meetings	The team spans multiple time zones (UK, Central Europe and Eastern Time); meetings are scheduled within the overlap, and anyone unable to attend receives minutes, recording and contributes asynchronously beforehand. Meetings occur twice weekly (the 2nd checkpoint being an optional drop-in).
WhatsApp	Group chat for day-to-day coordination. Responses expected within 24 hours on weekdays and review of a teammate's draft within 48 hours.
GitHub	For the team to work within day-to-day exclusively for code.
Google Drive	One shared drive folder for all project files.

Table 1: Communication Channels

3. Project scope and plan

Scope: The project covers HSBC and Barclays across ten consecutive reporting quarters, providing 2.5 years of longitudinal data for within-bank temporal analysis and same-quarter peer comparison. Within that corpus we separate the management presentation from the analyst question-and-answer session, attribute speaker turns, extract numeric claims, measure topic and sentiment, map topics to prudential risk categories, and produce a comparable state summary for each bank-quarter. Excluded are non-public regulatory data, audio and webcast processing, macroeconomic forecasting, historical stress-scenario modelling, agentic external search, live streaming analysis and model distillation. We also exclude any conclusion about either firm's actual prudential condition: the deliverable is a method and the evidence it produces.

Objectives:

Objective ID	Detail
O1	A reproducible evidence pipeline producing a provenance-preserving record of every speaker turn and question-and-answer pair (allowing for us to retain a complete and traceable understanding of the data) across both banks, by 19 September .
O2	A validated set of conversational and behavioural signals per question-and-answer pair, by 30 September .
O3	Taxonomy defined by 18 September ; topics mapped to it and dual-coded, by 30 September .
O4	Evidenced comparative findings, recommendations and stated limitations, delivered as a report and a fifteen-minute presentation, by 12 October .

Table 2: The project objectives

Metrics:

After a very preliminary investigation, within the two banks' datasets, for themes related to 4 topic areas (Profitability; Efficiency; Asset Quality/ Risk; Capital Adequacy) the themes mentioned were reviewed in order to decide upon an initial set of metrics and measures for the implementation phase. These are summarised in *Table 3* and *Table 4* below. Caveat: Once we begin the lower level analysis in the implementation phase we may further amend the list of measures and metrics if the investigation begins to reveal additional insightful signals into the firms' prudential conditions.

Topic	Mentioned Themes	Metric	Measures
Profitability	Revenue stability, structural hedge tailwinds, and net interest margins	Revenue stability, structural hedge tailwinds, and net interest margins	Group RoTE > 12% BUK RoTE > 20% USCB RoTE > 12%
Efficiency	Platform modernization, tech productivity, Structural Cost Actions (SCAs), and support function optimization	Cost-to-Income Ratio (CIR %), Gross Efficiency Savings (£bn), SCA Spend (£m)	Cost-to-Income Ratio: 61% FY25, high-50s 2026, low-50s 2028
Asset Quality/ Risk	Credit risk monitoring, US card loan losses, private credit exposures, delinquency trends	Cost of Risk / Loan Loss Rate (bps), Delinquencies (%), Value at Risk	Group Loan Loss Rate: 50–60bps USCB Loan Loss Rate: ~500bps
Capital Adequacy	Capital generation, shareholder buybacks, progressive dividends, Pillar 2A, and Basel 3.1	CET1 Ratio (%), Risk-Weighted Assets (RWAs £bn), Total Distributions (£bn)	CET1 Ratio: 13%–14% target range Total distributions: > £15bn

Table 3: Key Topic-related Themes mentioned in Barclays Bank's transcripts (after very preliminary investigation) in order to identify metrics and measures in scope for the investigative analysis

Topic	Mentioned Themes	Measures	Recommended Measure
Profitability	Total income revenue, return to equity, customer growth	Income growth, NII, fees, new to bank customers, group revenues, profit before tax	NII, New to bank customers, income growth
Efficiency	Operating costs, Exits	Cost/Efficiency language, cost growth, redeployment (how many branches closed)	Number of branches closed, cost growth, profit margins
Asset Quality/ Risk	Credit impairment (ECL), Uncertainty	Impairment, cost of risk, FX, deposits and loans, commercial real estate, LTV, interest rate (Feds)	FX, Cost of risk, NPL Ratio (non-performing loans)
Capital Adequacy	CET1, Deposits	Capital discussion, redeployments, exits, interest rates	CET1 change, LCR (Liquidity coverage ratio), LTD (loan to deposit ratio)

Table 4: Key Topic-related Themes mentioned in HSBC's transcripts (after very preliminary investigation) in order to identify metrics and measures in scope for the investigative analysis

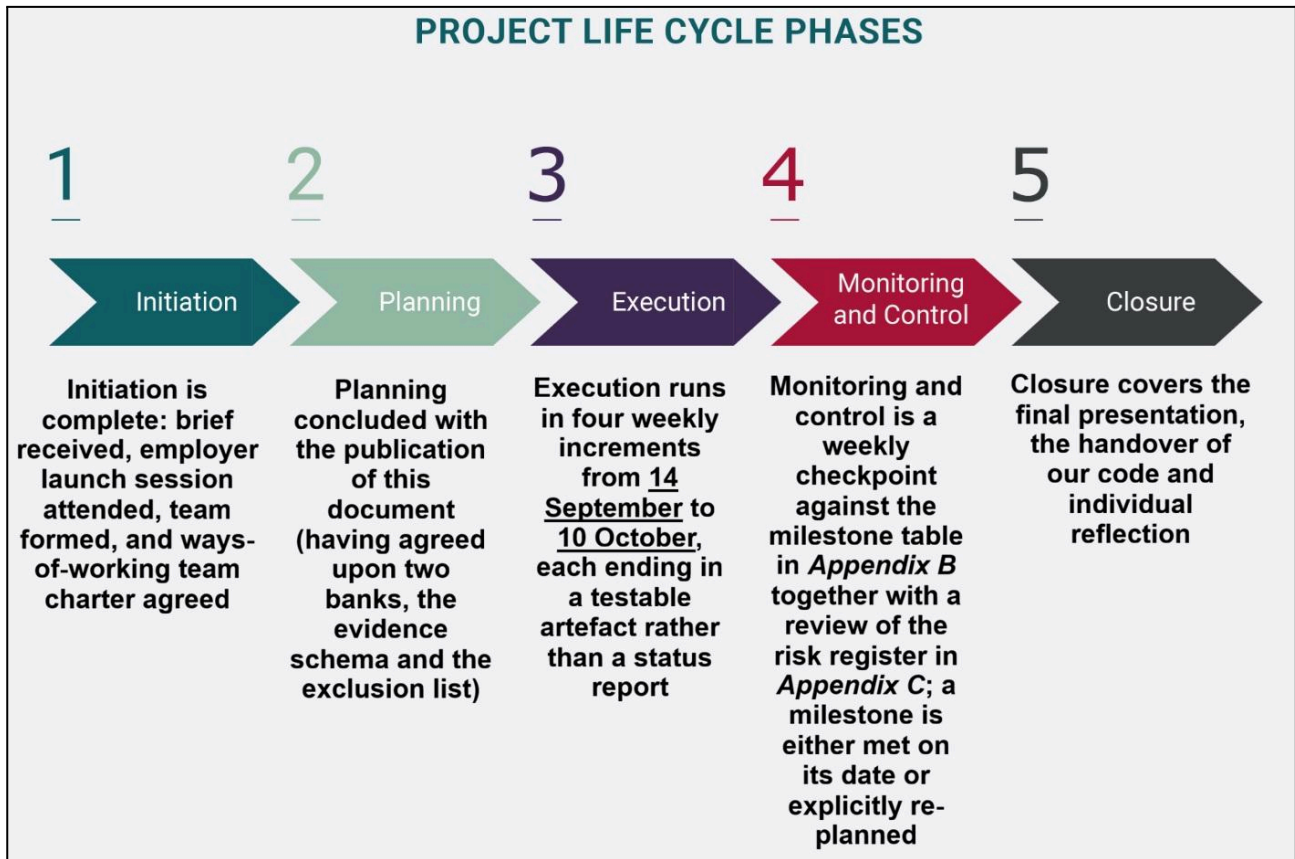
Life cycle:

Figure 1: The project's lifecycle phases outlined

4. Approach and work plan

We begin with a deterministic baseline, a keyword taxonomy derived from the prudential risk categories, so that a result exists independently of any model. We then evaluate pre-trained models against that baseline: a topic model over analyst questions, and a finance-domain sentiment model applied separately to questions and to answers. Rather than fine-tuning from scratch, which our 6-week timeline does not allow for, we evaluate first and adapt only where measured performance justifies it; any adaptation will be lightweight and documented. Integration is through files, each stage writing a defined artefact that the next stage consumes, making the pipeline re-runnable. Comparisons aggregate before they label, so the grouping level is a parameter rather than a hard-coded column: the same code produces a view by raw topic or by prudential category, keeping M7's fallback a configuration change.

Evaluation is explicit, as we validate a stratified sample of at least sixty question-and-answer pairs against human labels and report agreement rates rather than asserting accuracy.

Each phase is approximately one week (with ~2days of buffer before each submission):

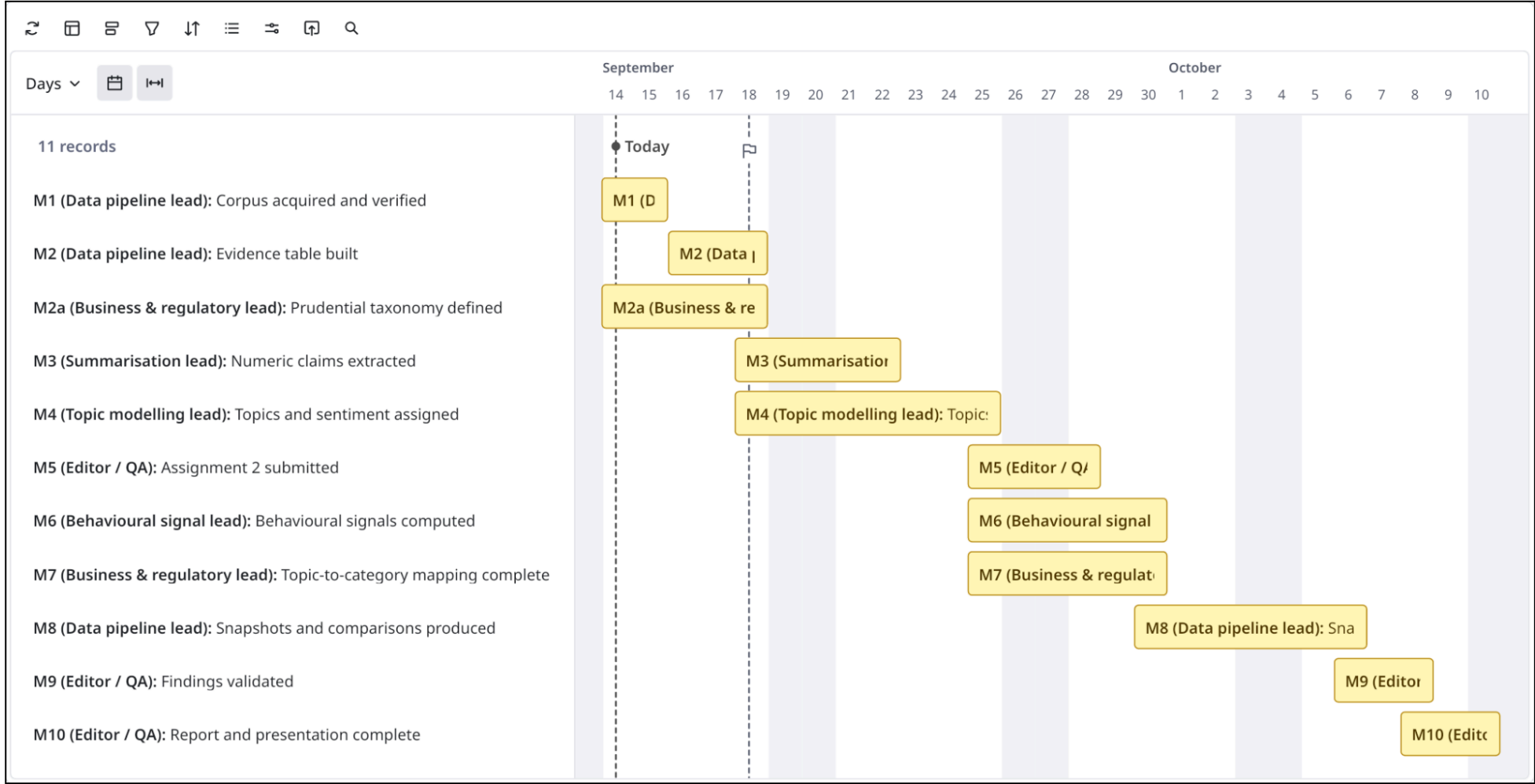
- Evidence extraction to **19 September**
- Signal measurement to **30 September**

- Supervisory mapping to **30 September**
- Comparison and validation to **8 October**

Key milestones, the test that determines whether each has been met, and the fallback if one is blocked are set out in *Appendix B*.

Appendix A: Project roadmap

The overarching milestone-centric roadmap is shown below.



Appendix B: Work plan and measurable milestones

Each milestone states the test by which it is judged complete, and the action taken if it is blocked.

ID	Date	Milestone and its test	Owner	If blocked
M1	Tue 15 Sep	Corpus acquired and verified. Manifest lists 20 documents (2 banks × 10 quarters) with zero missing quarters and a recorded source and publication date for each.	Data pipeline lead	Missing quarter: reduce to eight quarters per bank, log the decision, proceed the same day.
M2	Fri 18 Sep	Evidence table built. Turns and question-answer pairs extracted for every document; at least 95% of turns attributed to a named speaker and speaker type; every row carries bank, quarter, date, section and provenance.	Data pipeline lead	Layout variance between banks: build a parser per bank; two days of buffer held in week 3.
M2a	Fri 18 Sep	Prudential taxonomy defined. Eight risk categories documented from public PRA material, each with a definition and seed keywords, agreed by the team.	Business & regulatory lead	Depends only on public material, so it can start immediately. If delayed, adopt a reduced four-category scheme to unblock the M4 fallback and expand it later.
M3	Tue 22 Sep	Numeric claims extracted, each with value, unit and time horizon; precision of at least 0.8 measured on a 50-row manual audit.	Summarisation lead	Below threshold: restrict to a defined metric list rather than open extraction.
M4	Fri 25 Sep	Topics and sentiment assigned. Every question-answer pair carries a topic label and separate question and answer sentiment; at least 70% agreement with human labels on a 60-pair stratified sample.	Topic modelling lead	Unstable clusters: fall back to the seeded keyword taxonomy. Decision point fixed on this date.
M5	Mon 28 Sep	Assignment 2 submitted: deck as PDF plus a recording of 15 minutes ±10%.	Editor / QA	Dry run held 26 September so re-recording remains possible.
M6	Wed 30 Sep	Behavioural signals computed. Answer directness, metric-request coverage and topic-substitution rate calculated per pair and aggregated per bank-quarter.	Behavioural signal lead	Reduce to directness and metric coverage; substitution rate is the droppable measure.
M7	Wed 30 Sep	Topic-to-category mapping complete. Every observed topic mapped to one of the eight categories defined in M2a, dual-coded by two members with disagreements resolved and recorded. Depends on M4.	Business & regulatory lead	Produce M8's comparisons descriptively on raw topic clusters and apply the category labels afterwards; the comparison code groups by a parameter, so the switch is a configuration change rather than a rebuild.

ID	Date	Milestone and its test	Owner	If blocked
M8	Tue 6 Oct	Snapshots and comparisons produced. One state summary per bank-quarter plus all three comparison views, rendered with axes and sample sizes labelled. Consumes the mapping from M7.	Data pipeline lead	Drop the peer comparison first; the other two carry the argument alone.
M9	Thu 8 Oct	Findings validated. Every headline finding traced to at least three evidence records by identifier; limitations section written and agreed by the team.	Editor / QA	A finding that cannot be traced is cut rather than softened.
M10	Sat 10 Oct	Report and presentation complete. Report at 1,500 words $\pm 10\%$, deck timed at 15 minutes $\pm 10\%$ in a full dry run, code runnable end to end from the artefact files.	Editor / QA	Two days of buffer to the deadline by design.
—	Mon 12 Oct	Assignment 3 submitted: report, code, deck and live presentation to the Bank of England.	Coordinator	—

Appendix C: Risk register

Risks are reviewed at the weekly checkpoint described in section 3.

ID	Risk	Mitigation	Owner
R1	A bank does not publish a usable question-and-answer transcript for some quarters, or changes its document layout mid-series.	Verify the full series before any parsing work begins (M1). Build a parser per bank rather than one generic parser. If a quarter is missing, reduce to eight quarters per bank and record the decision.	Data pipeline lead
R2	The topic model produces unstable or uninterpretable clusters, stalling everything downstream.	Fall back to the seeded keyword taxonomy delivered by M2a on 18 September, which guarantees an interpretable result. The decision point is fixed at 25 September, matching M4, so the fallback still leaves time.	Topic modelling lead
R3	The sentiment model misreads hedged, heavily reviewed management language, producing confident but incorrect signals.	Human-validate a stratified sample of at least sixty pairs, report agreement rates, and state the limitation rather than concealing it.	Sentiment lead
R4	Divergence between the two banks reflects different business models rather than different risk.	Compare each bank against its own history first and treat cross-bank differences as questions to investigate rather than findings. State the confound explicitly in the report.	Business & regulatory lead
R5	Scope creep towards forecasting, scenario modelling or agentic components.	The exclusion list is fixed at this assignment. Any addition requires a written decision from the whole team and must displace something already in scope.	Coordinator
R6	A component owner becomes unavailable at a critical point.	The two critical components (Data pipeline and Summarisation) are owned by pairs. The remaining components (e.g. Topic modelling) will be areas that the rest of the team will need to 'swarm' on should the owner become unavailable - and all work lives in shared files rather than personal notebooks.	Coordinator
R7	Compute or environment limits interrupt long model runs.	Cache intermediate artefacts as files after every stage so a failed run resumes rather than restarts. Full-corpus runs are scheduled off the critical path.	Data pipeline lead

Appendix D: Roles and responsibilities breakdown

Role	Owner	Responsibility	Drawing on Strengths	Involved
Coordinator	Tasneem (Taz) Ahmad	Leads delivery, cadence, agendas, minutes, keeping the roadmap current; milestone tracking, and contact with our tutor and our BoE volunteer.	Given their experience of delivery management & team coordination	-
Data pipeline lead	Susana Venda	Code base setup, technical lead & technical coordination (including the parameterised grouping convention shared across comparison outputs), sourcing raw data from in-scope banks, and preprocessing transcripts.	Building on their strong technical leadership skills, and studied knowledge of data science techniques & principles	Rafael
Topic modelling lead	Debanjan Mukhopadhyay	Leading the topic modeling element of the pipeline using BERTopic to group documents into clusters to discover hidden themes.	Building on prior applied experience	-
Sentiment lead	Qianyi (Alfred) Sun	Leading the NLP element of the pipeline using FinBERT to score/ classify the data by sentiment.	Building on prior sentiment-vs-market educational experience	-
Behavioural signal lead	Bupathi Sankar Muthukumar	Leading the behavioural measurement element of the pipeline: answer directness, metric-request coverage and topic substitution per Q&A pair, aggregated into a per-bank-quarter series. Owns the definition of each measure and the hand-labelled sample used to validate it.	Building on their data analysis experience and comfort constructing derived features from structured records.	Sentiment lead (Alfred) and Summarisation lead (Rafael)
Summarisation lead	Rafael Navas Gascón	LLM summarisation pipeline will likely be a pivotal part of our analysis, as the finding summaries have to be grouped in a manner that makes sense (e.g. split into meaningful units such as Q&A pairs, enriched with labels for units, and grouped).	Building on their strong engineering & data categorization experience, comfort with messy text and structural thinking, as well as their patience.	Bupathi
Business & regulatory lead	Aidan Cameron	Conducts market research (e.g. Porter's Five Forces), and builds the prudential risk taxonomy from public material and interprets what each measured signal would mean to a supervisor.	Given their financial & banking experience making them an invaluable SME and supervisory & domain analyst	-
Editor / QA	Tasneem (Taz) Ahmad	Owns synthesis, validation and communication: assembling results, tracing every finding back to its evidence, and producing the report and presentation.	Given their strong consultancy skills, prior Quality Assurance and Business Analyst experience	Aidan

Appendix E: Team's PREACH collaboration rules

PREACH framework Area	PREACH framework Detail	Team's related collaboration rule agreement
P	Project Management and Communication	What communication tools or channels will you use to collaborate and manage your project? The WhatsApp group chat for day-to-day (quick questions, blockers, availability changes. Video calls for anything needing a decision. One shared drive folder for all project files. Git repo for the team to work within day-to-day for code exclusively. Email only for anything that has to reach the tutor or success manager.
		How will you manage team workload & assign tasks? Every deliverable has one named owner and a date, visible in one place all seven of us can see — a shared board or sheet, agreed at the kickoff. Work is split by section or component, not pooled: pooled work with seven people means the same paragraph gets written three times and the hard part gets written by nobody
R	Relationships and Team roles	Who is the project lead, defined here as the person who will coordinate the project? Will this role rotate at certain intervals? Taz initially. Potentially rotate the leadership role, based on phase of project and interest from other team members.
		What roles will other team members undertake? Think about your technical strengths and your human strengths, do you excel at ideation or are you amazing at proofreading? Alfred: (1) Strengths - Modeling experience with academic projects. (2) Specialist, Plant, Shaper. Taz: Coordinator with some Implementer. I'd rather someone stronger on the modeling side owns the technical approach.
E	Empathy and Listening	How will you make sure all members of the team are heard? Every meeting ends with a round where each person speaks - the quietest voice in a seven-person call is usually the one that spotted the problem. Decisions get written into the minutes with the reasoning, so anyone who couldn't attend can push back afterwards rather than discovering it later.
		How will you reconcile different ideas, points of view, or disagreements? Group majority (e.g. WhatsApp poll) to get a consensus. If there's still no consensus, then time-box it - 5-10 minutes of discussion, max. Then the person who owns that component decides and we record the alternative in the minutes. If it changes scope, it comes to the whole team instead.
A	Attitude and Availability	Choose three attitudes or values that define your team ethos. Communicative, Collaborative, & Innovative

		• We aim to keep communication channels effective and open • We aim to lean-in when teammates need support/ a sounding board for the project • We want to utilize our learnings as well as bring our authentic selves to the table in order to solve problems/ find insights that haven't been surfaced before.
		When are meetings held and what communication is expected outside of meetings? Meetings: Two calls a week plus async updates (1) Saturday mandatory (12pm UK/ 1pm CEST/ 7am EDT); (2) Sunday optional (12pm UK/ 1pm CEST/ 7am EDT); Communication outside of meetings: (1) Weekdays WhatsApp responses: 24-hour response norm on weekdays (2) Weekend WhatsApp responses: 4-hour responses on weekends between 9am - 9pm local timezone of respondent.
C	Contribution and Escalation	What defines satisfactory contribution in your team? You deliver what you own by the agreed date, or you flag at least 24 hours ahead that you can't; you attend the agreed meetings or send your input in advance; and you review other people's drafts within 48 hours of being asked.
		If contribution is lacking how will you address and escalate this? If contribution falls short a direct, private conversation first, framed with Situation–Behaviour–Impact - what happened, what the effect was, no character judgments. If it continues, it's raised with the whole team so the work can be redistributed while there's still time. If it still isn't resolved, we raise it in the Cambridge Tutor session/ Success Team. Nobody gets escalated without having had/ attempted the direct conversation first.
H	Help and Feedback	How will you support your teammates? Ask early. A blocker raised on day one is a fifteen-minute conversation; the same blocker raised the night before a deadline is someone else's weekend. If you're stuck, say so in the chat and one of us pairs with you. Whatever component something goes wrong in, the emphasis should be on the component leader to keep track of progress and to tackle blockers.
		How and when will you give feedback on tasks completed or work in progress? Feedback: • Github for code reviews. • Google doc comments for document related feedback. On drafts within 48 hours, and at the milestone rather than at the end - feedback on a final draft the day before submission isn't feedback, it's a problem. Same SBI structure for anything about how we're working, not just what we produced.